

6 Concept Scrubbing

Unfortunately, Elazar et al. [10] were forced to limit their interventions to a single layer due to the limitations of INLP. INLP often requires the deletion of several dozen dimensions before linear guarding is achieved—as demonstrated in Figure 2. Kumar et al. [21] show empirically and theoretically that INLP causes needless “collateral damage” to useful parts of the embedding that are orthogonal to the concept being erased. Because of this collateral damage, it’s impossible to apply INLP to multiple layers of a transformer without causing its outputs to collapse into gibberish.

Instead, we would like to erase all linear information about a concept in the activations at *every* layer, which we term **concept scrubbing**. LEACE makes concept scrubbing possible and eminently practical. It causes minimal collateral damage, induces little computational overhead, and the covariance statistics it relies on can be computed in a *streaming* fashion, without ever storing all the hidden states in memory or on disk.

Algorithm. Any intervention on the model at layer ℓ changes the distribution of hidden states at layers $\ell' > \ell$. Because of this, the naive approach of independently fitting LEACE parameters $(\mathbf{P}, \mathbf{b})$ for all layers of the clean model, then applying them all at once, may fail to fully erase the target concept. Instead, we fit LEACE parameters *sequentially*, starting from the first layer and proceeding to the final layer. After we compute $(\mathbf{P}, \mathbf{b})$ for a layer, we immediately use them to scrub the hidden states for that layer, then feed these scrubbed embeddings to the next layer (Algorithm 1).

Algorithm 1 Concept scrubbing

Require: Model with ℓ layers $f = f_\ell \circ \dots \circ f_1$
Require: Design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$
Require: Label matrix $\mathbf{Z} \in \mathbb{R}^{n \times k}$
Ensure: LEACE parameters for each layer in f

- 1: $\mathbf{H}_1 \leftarrow \text{Embed}(\mathbf{X})$
- 2: $L \leftarrow \text{list}()$
- 3: **for** $l \in 1 \dots \ell$ **do**
- 4: Fit $(\mathbf{P}, \mathbf{b})$ on $\mathbf{H}_l$ and $\mathbf{Z}$
- 5: Append $(\mathbf{P}, \mathbf{b})$ to L
- 6: $\mathbf{H}_l \leftarrow \mathbf{P}(\mathbf{H}_l - \mu_{\mathbf{H}_l}) + \mu_{\mathbf{H}_l}$ (Eq. 1)
- 7: $\mathbf{H}_{l+1} \leftarrow f_l(\mathbf{H}_l)$
- 8: **return** L

Condition	LLaMA			Pythia			
	7B	13B	30B	160M	1.4B	6.9B	12B
No intervention	0.69	0.66	0.62	0.90	0.70	0.64	0.62
Random erasure	0.69	0.66	0.62	0.99	0.72	0.66	0.63
LEACE	1.73	1.84	1.96	2.79	2.25	3.57	3.20
SAL	3.24	3.26	3.16	3.53	3.44	4.17	4.69
unigram entropy	2.90	2.90	2.90	2.66	2.66	2.66	2.66

Table 1: Perplexity in autoregressive language models when removing linearly available part-of-speech information from the input to each transformer layer. Units are bits per UTF-8 byte. The unigram baseline assigns probabilities to tokens based only on their frequency and not on the context.

6.1 Experimental Details

Dataset. For each model family, we use a sample from the respective pretraining distribution: the validation split of the Pile [13] for the Pythia models [2], and the RedPajama replication of the LLaMA pretraining corpus for the LLaMA family [39]. sampling a slice of 2^{22} tokens for fitting the LEACE parameters and another slice of 2^{22} tokens for evaluation. Since neither corpus comes with part-of-speech tags, we use the model from the SpaCy library [19] to automatically generate Universal Dependency tags [23].

Baseline method. We also run concept scrubbing using full-rank SAL [36], which is similar to our method but lacks a bias term and does not adjust for correlations between features (Appendix D).

Architecture. We focus on autoregressive language models. We evaluate our method on EleutherAI’s Pythia 160M, 1.4B, 6.9B, and 12B models [2], and Meta’s LLaMA 7B, 13B, and 30B [39]. We apply concept erasure to the input of each transformer block, immediately after normalization is applied (LayerNorm or RMSNorm).

Randomized erasure. Almost any intervention on a neural network will cause its performance to degrade to some extent. Following Elazar et al. [10], we isolate the effect of the concept erasure by comparing it to a control condition in which we orthogonally project onto a *random* linear subspace of the same rank as the cross-covariance matrix. To reduce the variance of our results, we sample a fresh subspace for each minibatch, and erase that subspace at each layer, reporting the cross-entropy loss averaged over subspaces.

Training efficiency. Algorithm 1 avoids redundant computation by caching the layer i hidden states for *every* data point, then using them to run layer $i + 1$. This approach has the downside of requiring a large amount of memory or disk space during training (up to 500GB in our experiments). It’s possible to avoid caching any hidden states and instead recompute them as needed, at the expense of increasing the total compute cost from $O(\ell)$ to $O(\ell^2)$.

6.2 Results

We find strong evidence that autoregressive language models heavily rely on linearly encoded part-of-speech information. While erasing a randomly selected subspace has little to no effect on language modeling performance, scrubbing away part-of-speech information induces a large increase in perplexity across all models (Table 1).

The specific numbers, however, depend on the erasure method used: SAL induces significantly larger increases in perplexity for all models we tested. We take this to mean that SAL inflicts more collateral damage on other useful features in the embedding than LEACE does. In other words, interventions made with LEACE are more *surgical* than those made with prior work; they more closely approximate the ideal of a perfect intervention which only erases the target concept and keeps everything else fixed [40, 15]. If this experiment were conducted with SAL alone, we would have *overestimated* the causal effect of part-of-speech.

7 Limitations and Future Work

Much work remains to be done to validate concept scrubbing. Specifically, we’d like to see experiments that target concepts much narrower than part-of-speech, and use behavioral metrics to determine whether scrubbing changes the network in the ways we’d intuitively expect. If these experiments succeed, an exciting next step would be the incorporation of concept scrubbing into the pretraining and/or finetuning process. This may make it possible to train deep neural networks subject to *conceptual constraints*. It remains to be seen if gradient-based optimizers will be able to “circumvent” such constraints by encoding protected attributes in completely nonlinear ways.

In this work, we focused exclusively on *linear* concept erasure due to its simplicity and tractability. Some authors have proposed nonlinear concept erasure techniques based on kernel methods, but have found that erasure functions fit using one kernel do not generalize well to other kernels [32, 36]. We conjecture that it is intractable to nondestructively edit X so as to prevent a general nonlinear adversary from recovering Z , unless the data generating process for X is known in detail.⁶

A major motivation of concept erasure is that it promises to prevent models from using a concept in a *post hoc*, model-agnostic fashion. But if our concept scrubbing procedure turns out to yield unsatisfactory results in practical use cases, the most promising research direction might then be to improve model-*specific* techniques, such as those that modify the training procedure [8, 9, 14].

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⁶We suspect erasing a concept is at least as hard as extracting it from the original embedding. But in the worst case, information about Z could be encoded *cryptographically* in X , which would be intractable to decode given standard computational complexity assumptions. If the data is generated by a known algorithm, however, it may be possible to efficiently eliminate mutual information between Z and X by simply breaking the links in the causal graph that connect them.

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